Guidelines for the Description of Wheat Germplasm Resources

1 scope

This specification sets out the descriptors for wheat germplasm resources and their grading criteria.

These guidelines apply to the collection, organization, and preservation of wheat germplasm resources; the development of data standards and data quality control specifications; and the establishment of databases and information-sharing network systems.

2 Normative References

The provisions of the following documents shall form part of this specification by reference therein. For dated reference documents, all subsequent corrigenda (excluding errata) or revised editions shall not apply to this specification; however, parties to agreements concluded in accordance with this specification are encouraged to consider whether the latest editions of those documents may be used. For undated reference documents, the latest editions shall apply to this specification.

ISO 3166 Codes for the Representation of Country Names: GB/T 2659 – Country and Region Name Codes for Countries and Regions Worldwide

GB/T 2260: Administrative Division Codes of the People's Republic of China

GB/T 12404 Code for Unit Affiliation

GB/T 3543-1995 Code for Inspection of Crop Seeds

GB/T19557.2-2004 Guidelines for DUS Testing of New Wheat Varieties

3 Terms and Definitions

3.1 wheat

Gramineae: perennial or annual herbaceous plants belonging to the genus Triticum L. There are currently approximately 27 species worldwide; the most widely utilized in production is common wheat *Triticum aestivum L*.

3.2 Wheat Germplasm Resources

Plant materials that can be used to transfer germplasm to wheat include plants from all species within the genus *Triticum* and their closely related genera. These resources encompass both wild and cultivated species, local varieties, developed varieties, and introduced varieties; as well as lines with distinctive superior traits, mutants, male sterility lines, and aneuploids.

3.3 essential information

Basic descriptive information on wheat germplasm resources, including the nationally unified identification number, germplasm name, place of origin, storage institution, breeding institution, breeding year, and breeding method, among others.

3.4 Morphological characteristics and biological properties

The morphological characteristics and biological traits of wheat germplasm resources, as well as their economic traits and traits associated with agricultural activities. In practical applications, these include morphological traits (such as flag leaf morphology, husk color, spike shape, grain color, and grain shape), developmental traits (such as the jointing stage, heading stage, and maturity stage), yield-related traits (such as the number of grains per spike and 1,000-grain weight), and traits associated with agricultural activities (such as the sowing stage and emergence stage).

3.5 quality characteristic

The nutritional quality of wheat germplasm resources, including grain, comprises crude protein, lysine, wet gluten content, and sedimentation values, among other parameters.

3.6 Stress resistance

The adaptability or resistance of wheat germplasm resources to various abiotic stresses,

including drought tolerance, salt tolerance, cold tolerance, and moisture tolerance.

3.7 disease resistance

The resistance of wheat germplasm resources to the invasion and spread of disease-causing pathogens. Resistance to disease immunity refers to resistance to invasion; a high degree of resistance to disease indicates resistance to spread; and weak resistance indicates susceptibility to disease. The principal diseases affecting wheat germplasm resources include stripe rust, leaf rust, stem rust, powdery mildew, Fusarium head blight, root rot, damping-off, yellow dwarfing, and total blight.

3.8 insect resistance

The resistance of wheat germplasm resources to pest damage. The primary pests affecting wheat germplasm resources include aphids, phloem-sucking insects, and others.

3.9 Wheat growth cycle

Based on the characteristics of wheat at different growth stages, the entire life cycle of wheat can be divided into 12 growth periods: emergence, three-leaf stage, tillering, overwintering, regreening, stand-up, jointing, heading, flowering, grain filling, and maturity. Generally, winter wheat requires 230–270 days, while spring wheat requires 90–150 days. Based on the characteristics of wheat organ formation, several consecutive growth periods can be merged into a specific growth stage. The seedling stage, from emergence to the stand-up stage, is primarily characterized by vegetative growth, focusing on root elongation, leaf expansion, and tillering. The intermediate stage, from stand-up to flowering, is a phase where vegetative and reproductive growth occur simultaneously; this involves the growth of roots, stems, and leaves, as well as the differentiation and development of the wheat ear. The late stage, from flowering to maturity, is also known as the grain formation stage and is primarily characterized by reproductive growth.

4 essential information

4.1 Nationwide Unified Number

The unique identifier for germplasm: domestic common wheat germplasm resources are numbered as "ZM" followed by a six-digit sequential number; foreign common wheat germplasm resources are numbered as "MY" followed by a six-digit sequential number.

4.2 Germplasm Bank Identifier

The accession number of wheat germplasm in the National Long-term Crop Germplasm Resource Bank consists of "I1B" followed by a five-digit sequential number.

4.3 Introduction Number

The designation assigned to wheat germplasm when it is introduced from abroad.

4.4 Collection ID

The identification number assigned to wheat germplasm during field collection.

4.5 Germplasm name

The Chinese name for wheat germplasm.

4.6 Genetic resource foreign name

The foreign-language name of the introduced wheat germplasm, or the Pinyin name used for the international exchange of domestic germplasm.

4.7 Field Name

Gramineae.

4.8 generic name

The genus Triticum (*Triticum* L.)

4.9 Scientific Name

The scientific name of common wheat is *Triticum aestivum L.*

4.10 Country of Origin

The name of the country of origin, the name of the region, or the name of the international organization of the wheat germplasm resources.

4.11 Country of Origin

Name of the province of origin for domestic wheat germplasm; national first-level administrative region of origin for germplasm introduced from abroad

The name of the district.

4.12 Country of Origin

The names of the native counties, townships, and villages of domestic wheat germplasm.

4.13 Height

The elevation of the origin of wheat germplasm, expressed in m.

4.14 Longitude

The longitude of the origin of wheat germplasm is expressed in degrees (°) and minutes ('). The format is DDDFF, where DDD represents degrees and FF represents minutes.

4.15 Latitude

The latitude of the origin of wheat germplasm, expressed in degrees (°) and minutes ('). The format is DDFF, where DD represents degrees and FF represents minutes.

4.16 Origin

For wheat germplasm introduced from abroad, provide the name of the country, region, or international organization from which it originated; for domestic germplasm, provide the name of the province or county of origin.

4.17 Preserving entity

The name of the original institution where the wheat germplasm was preserved prior to its submission to the National Long-term Crop Germplasm Resource Bank.

4.18 Save Unit Number

The germplasm accession number assigned by the original germplasm preservation institution.

4.19 genealogy

The genetic relationships among wheat breeding varieties (lines).

4.20 Breeding Unit

The name of the institution or individual responsible for the breeding and development of wheat varieties (lines).

4.21 Age at maturation

Years in which wheat varieties (lines) were successfully developed.

4.22 Breeding Method

Breeding methods for the development of wheat varieties (lines).

4.23 Genetic Material Type

Wheat germplasm types are divided into six categories.

1 Wild resources

2 local variety

3 Selected breeding varieties

4 Strain

5 genetic stocks

6 other

4.24 picture

Image filename for wheat germplasm. The image format is.jpg.

4.25 observation place

Name of the observation site for the morphological characteristics and biological traits of wheat germplasm.

5 Morphological characteristics and biological properties

5.1 Winter–spring Wheat

Wheat is classified according to the sowing season into two types: winter wheat and spring wheat. Winter wheat refers to wheat varieties sown in autumn and harvested in the following summer; spring wheat refers to wheat varieties sown in spring and harvested during the summer and autumn of the same year.

1 winter wheat

2 spring wheat

5.2 Winter–spring Nature

The required duration and temperature level of the vernalization stage during the seedling stage of wheat germplasm.

1 winter

2 Weak winter

3 spring

4 facultative

5.3 sowing time

The field sowing date for wheat shall be expressed as "year month day" in the format "YYYYMMDD".

5.4 Period of Emergence

After wheat sowing, the date on which more than 50% of the seedlings in the entire region emerge above ground by 2–3 cm shall be expressed as "year month day" in the format "YYYYMMDD."

5.5 Youth Recovery Period

For winter-type materials sown in autumn in northern regions, the date on which the aboveground leaves dry out or lose their green color due to the cold during winter, and by which the old leaves regain their green color and new leaves emerge in the following spring, is expressed as "year month day" in the format "YYYYMMDD".

5.6 Shooting Period

In the entire district, 50% of the plants have achieved stem elongation of 3–4 cm, with the first internode extending 1.5–

A date of 2.0 cm. It should be expressed as "year month day" in the format "YYYYMMDD".

5.7 Heading Period

The date on which 50% of the panicles in the entire district extend beyond the leaf sheaths of the flag leaves. Expressed as "year month day" in the format "YYYYMMDD."

5.8 efflorescence

The date on which 50% of the plants in the entire district have their panicles in bloom or have exposed anthers. This should be expressed as "year month day," in the format "YYYYMMDD."

5.9 maturation period

The date on which the grain wax reaches maturity shall be expressed as "year month day" in the format "YYYYMMDD".

5.10 Mature Nature

Compared with mid-maturing varieties widely cultivated locally, the time of maturation is categorized as follows:

1 Very early

2 morning

3 China

4 evening

5 Extremely late

5.11 Whole reproductive period

The number of days elapsed from the date of sowing to the date of maturity. The unit is d.

5.12 Photoperiodic response characteristics

The response of wheat germplasm plants to light duration.

1 slow

2 secondary

3 sensitive

5.13 period of dormancy

Once seeds have matured, they require a period of post-maturation before they can germinate; this period is known as the dormancy period. The length of the seed dormancy period varies significantly among different varieties.

1 short

2 China

3 long

5.14 Bud sheath color

The color when the young shoot has emerged from the soil to a length of approximately 2 cm.

1 Green

2 purple

5.15 Seedling Habits

The leaf posture of wheat germplasm during the peak tillering stage (see Figure 1).

1 up-right

2 semi-prostrate

3 crawl



1 2 3

Figure 1. Seedling Habits

5.16 Seedling Color

The color of the seedlings.

1 pale green

2 Green

3 Dark Green

5.17 Long Seedling Leaves

Length of the seedling leaves, expressed in cm.

5.18 Broad Leaf Blade

Width of the seedling leaves. Unit: cm.

5.19 Leaf pubescence

Presence or absence of pubescence on the leaf surface.

0 not have

1 have

5.20 plant type

The distribution of the main stem and tiller stems after heading (see Figure 2).

1 compact

2 secondary

3 loose



Figure 2: Strain Type

5.21 Ye Zi

Leaf morphology, angle between the stem and leaves, and degree of spreading (see Figure 3).

1 straight and upright

2 Flat

3 Lower cloak



2 3

Figure 3: Leaf Morphology

5.22 flag leaf length

The first leaf below the crown is called the flag leaf; the length of the flag leaf from its base to its tip.

1 short

2 China

3 long

5.23 flag leaf width

The first leaf below the crown is called the flag leaf; the width of the middle portion of the flag leaf blade.

1 narrow

2 China

3 wide

5.24 Flag Leaf Angle

The angle between the flag leaf and the lower stem of the panicle (see Figure 4).

1 straight and upright

2 Flat

3 Lower cloak



Figure 4: Flag Leaf Angle

5.25 Leaf Color

The color of the leaf auricles: the protrusions extending from both sides of the base of the leaf are called leaf auricles.

1 Green

2 purple

5.26 Anther Color

The anther is a key component of the stamen; there are two colors of anthers.

1 Huang

2 purple

5.27 Sui Wax

From the flowering stage through the grain-filling stage: the presence and intensity of powdery waxy material on the surface of the panicle.

0 not have

1 light

2 repeat

5.28 Stem waxy layer

From flowering through the grain-filling stage, the presence and intensity of powdery waxy material on the stem surface.

0 not have

1 light

2 repeat

5.29 wax layer

From flowering through the grain-filling stage, the presence and intensity of powdery waxy material on the leaf surface.

0 not have

1 light

2 repeat

5.30 spike shape

The shape of the mature panicles (see Figure 5).

1 Spindle

2 rectangular

3 circular cone

4 Rod

5 oval

6 branch

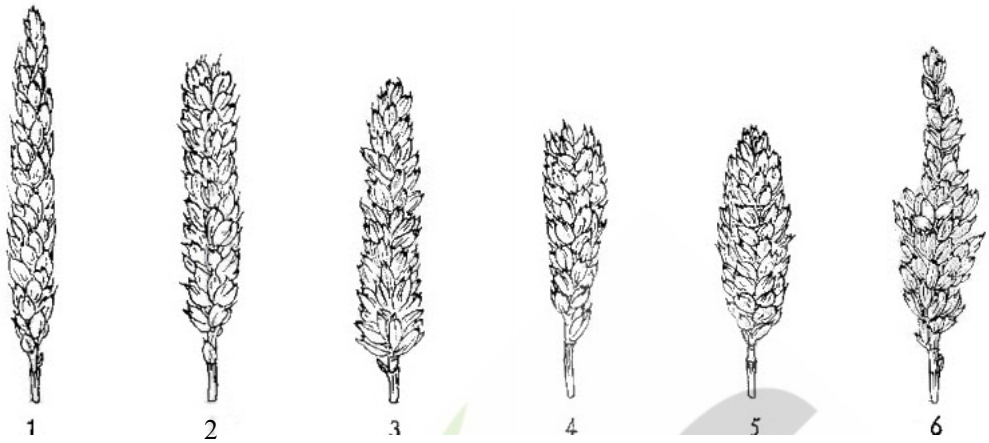


Figure 5 Spike-Shaped

5.31 Stem Color

Stem color during the waxing stage.

1 Huang

2 purple

5.32 Mang Shape

The presence or absence, length, and degree of curvature of the outer bulge tip extension (see Figure 6).

0 not have

1 short

2 long

3 Curvature

4 Short tune

5 Long melody

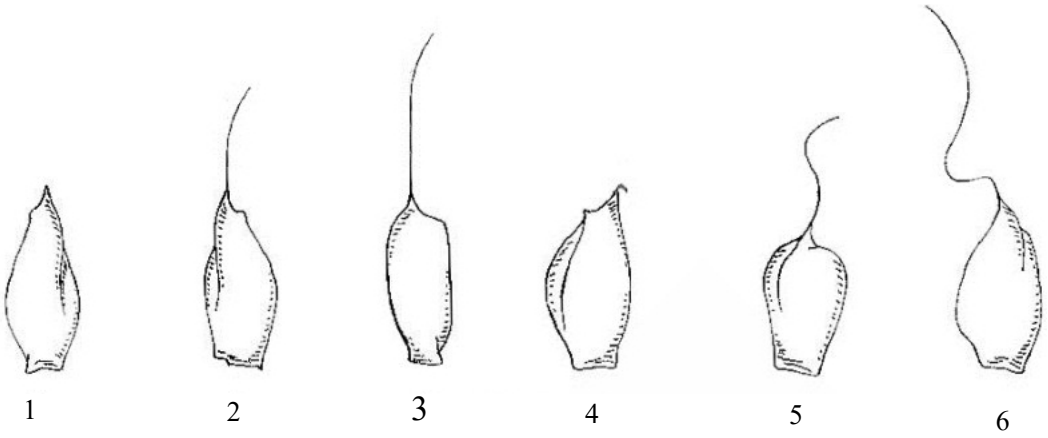


Figure 6:芒形

5.33 Mang color

When mature, the color of the awn is generally consistent with the color of the husk. Under normal circumstances, the awn color is not recorded; instead, only terms such as "black awn" are used.

5.34 Shell Color

The color of the perigamy and outer peristome during the mature stage.

1 white

2 red

3 black

4 White background with black patterns (edges)

5 Red background with black patterns (edges)

5.35 Shell Hair

Indicate whether the glume is pubescent; record only those glumes that are pubescent.

0 not have

1 have

5.36 Protective Tip Shape

The shape of the pericarp on the side is determined by the pericarp at the middle of the main spike (see Figure 7).

1 oval in shape

2 oval

3 ovum

4 rectangular

5 circle

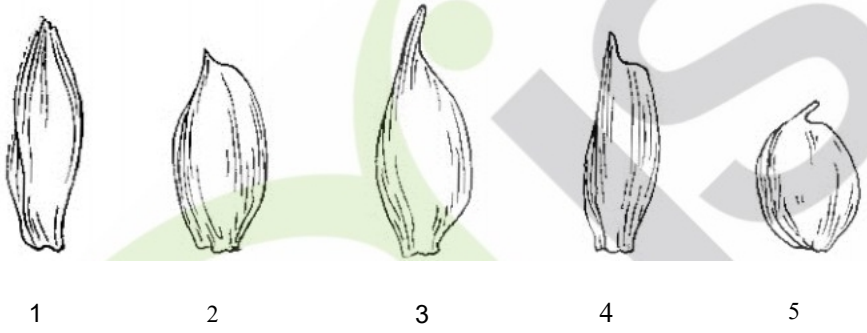


Figure 7: Shape of the Sheath

5.37 Ying Shoulder

The shape of the perigamy is determined by the perigamy at the middle of the main spike (see Figure 8).

0 not have

1 oblique

2 Fang

3 Hill

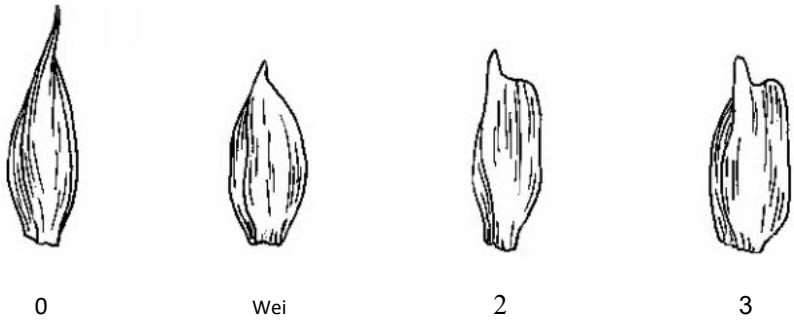


Figure 8: Ying-Shoulder

5.38 Yingzui

The shape of the glume at the tip is determined by that of the glume at the middle part of the main spike (see Figure 9).

1 blunt

2 Sharp

3 beak

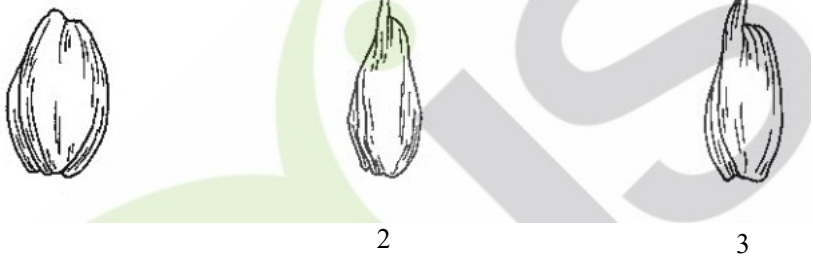


Figure 9: Yingzui

5.39 Yingji

The keel-like prominence in the middle of the perigamy is termed the perigamy ridge; in some varieties, the ridge bears serrations (see Figure 10).

1 non constat

2 clear

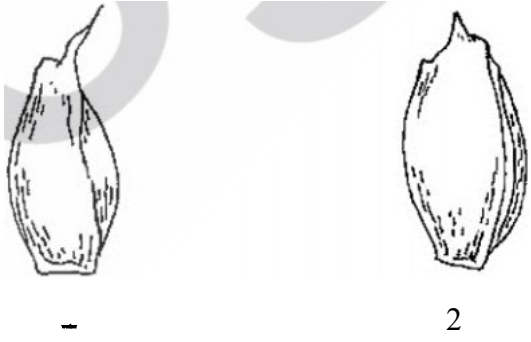


Figure 10 Ying Ridge

5.40 form of grain

The shape of the grains after complete maturation (see Figure 11).

1 oval in shape

2 ovum

3 oval

4 circle



Figure 11: Granule Morphology

5.41 ventral groove

The depth and width of the concave grooves on the ventral surface of the grains.

1 shallow

2 deep

5.42 Pappus

The density of trichomes at the apex of the grains (see Figure 12).



2

Graph 12 Pappus

5.43 Granule color

Color of mature grains.

1 white

2 red

3 Black-purple

4 blue or green

5.44 granular substance

The firmness of the endosperm of the grain after complete maturation.

1 soft

2 semihard

3 hard

5.45 **particle** size

Fully mature grains are classified by size according to their hundred-weight.

1 small

2 China

3 big

4 especially big

5.46 plumpness

The firmness of fully mature grains.

1 Not full

2 secondary

3 Plump

5.47 Grain uniformity

Assess the degree of size variation among the grains and visually determine whether they are uniform.

1 irregularity

2 secondary

3 Qi

5.48 plant height

Plant height: the length measured from the ground to the top of the panicle (excluding the awn) before and after the milk stage. Unit: cm.

5.49 Plant Uniformity

During the period from booting to maturity, the plant height and the uniformity of the heights of the main panicle and tiller panicles.

1 irregularity

2 secondary

3 Qi

5.50 tiller number

Total tiller number per plant, including viable tillers and nonviable tillers. Unit: number of tillers.

5.51 Effective Tiller Number

Any tiller that emerges and produces fruit is considered an effective tiller; count the number of effective tillers per plant. The unit is "number of tillers."

5.52 穗 Length

The length of the spike, measured from the base of the spike to its tip (excluding the awn). Unit: cm.

5.53 Density of spikelet arrangement

The average number of spikelets per 10 cm of the spike axis.

1 rare

2 China

3 secret

4 Extremely dense

5.54 Number of florets per spike

The total number of spikelets per spike, including sterile spikelets; the unit is [number] units.

5.55 Number of Sterile Spikelets

The number of florets at the lower and upper parts of a spike that are not fertile; the unit is the number of florets.

5.56 Number of Spikelets

Number of grains set in the spikelets bearing the greatest number of grains in the middle part of the spike; expressed in grains.

5.57 Number of Grains per Spike

Total number of grains per spike. Unit: grains.

5.58 Grain Weight per 100 G

Weight of grains per spike. Unit: g.

5. 59 kg per 1,000 grains

Weight of 1,000 dried seeds, expressed in g.

5.60 Biomass Yield per Plant

Including the weight of the roots, stems, leaves, and spikes. The unit is g.

5.61 seed holding

After the grains have fully matured, the degree to which they fall when exposed to strong winds.

1 Loose mouth

2 secondary

3 closemouthed

5.62 Lodging Resistance

The degree of plant recovery after lodging caused by heavy wind or rain from the heading stage through the maturity stage of wheat.

1 Strong

2 China

3 weak

4 Extremely weak

6 quality characteristic

6.1 seed moisture content

The mass of moisture contained in the seed as a percentage of the total seed mass, expressed as%.

6.2 unit weight

Mass of wheat grains per unit volume. The unit is g/L.

6.3 Hardness

The hardness or softness of wheat grains is expressed using the grinding time method (GT), with the unit being s.

6.4 Crude protein content

The ratio of crude protein content in the grain to the mass of the grain, expressed as a percentage (%).

6.5 Lysine Content

The ratio of lysine content in the grain to the mass of the grain, expressed as a percentage (%).

6.6 Settlement value

Within a standard time interval, the volume of flocculent precipitate formed by suspending flour in a sodium dodecyl sulfate–lactic acid solution at a specified concentration; the unit is ml.

6.7 Moist Gluten Content

The wet gluten content of wheat flour with a moisture content of 14%, expressed as%.

7 Stress Resistance

7.1 Drought Resistance During the Budding Stage

The ability of wheat germplasm to resist water stress during the seed germination stage.

1 Very strong (HR)
2 Strong (R)
3 Medium (MR)
4 weak(S)
5 Very weak (HS)

7.2 Seedling stage drought resistance

The ability of wheat germplasm to withstand soil drought or atmospheric drought during the seedling stage.

1 Very strong (HR)
2 Strong (R)
3 Medium (MR)
4 weak(S)
5 Very weak (HS)

7.3 Drought resistance throughout the entire growth period

Refers to the ability of wheat plants to withstand soil drought or atmospheric drought throughout their entire growth period.

1 Very strong (HR)

2 Strong (R)

3 Medium (MR)

4 weak(S)

5 Very weak (HS)

7.4 Salt tolerance during the budding stage

The tolerance of wheat to high concentrations of saline stress in the soil during the boot stage.

1 High endurance (HR)

2 Salt tolerance (R)

3 Medium Resistance (MR)

4 sensitive(S)

5 High Sensitivity (HS)

7.5 Salt tolerance during the seedling stage

The tolerance of wheat seedlings to high concentrations of saline stress in the soil during the seedling stage.

1 High endurance (HR)

2 Salt tolerance (R)

3 Medium Resistance (MR)

4 sensitive(S)

5 High Sensitivity (HS)

7.6 Salt tolerance throughout the entire growth cycle

The tolerance of wheat plants to high concentrations of saline stress in the soil throughout their entire growth cycle.

1 High endurance (HR)

2 Salt tolerance (R)

3 Medium Resistance (MR)

4 sensitive(S)

5 High Sensitivity (HS)

7.7 winter resistance

The ability of winter wheat seedlings in the north to withstand cold during winter.

1 Very strong (HR)

3 Strong (R)

5 Medium (MR)

7 weak(S)

9 Very weak (HS)

7.8 moisture-proof

Wheat's tolerance to soil moisture stress caused by excessive soil moisture.

1 Strong

2 China

3 weak

7.9 Anti-boot germination

Upon entering the maturity stage, during continuous rainy weather in the field or after harvest, the ability of the grains to resist germination on the panicles is affected.

1 highly resistance(HR)

3 resist(R)

5 moderate resistance(MR)

7 sensitive(S)

9 High Sensitivity (HS)

8 Resistance to pests and diseases

8.1 Barren Rust Resistance

The resistance of wheat plants to infection by the stripe rust pathogen (*Puccinia striiformis* Westend.)

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.2 Resistance to leaf rust

Wheat plants exhibit strong resistance to infection by the leaf rust pathogen *(Puccinia triticina* Eriks)

weak.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.3 Resistance to stalk rust

Resistance of wheat plants to infection by the stem rust pathogen (Puccinia *graminis* Pers. Pers.)

Strength and weakness.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.4 powder mildew resistance

Wheat plants' susceptibility to the powdery mildew pathogen (*Blumeria graminis (* DC.) E.O. Speer) infection

The degree of resistance caused by the pathogen.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.5 Fusarium head rot resistance

The resistance of wheat plants to infection by Fusarium fungi (*Fusarium* spp.).

1 highly resistance(HR)

3 disease-resistant(R)

5 Zhong Kang (MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.6 Root rot resistance

Wheat plants are susceptible to root rot pathogens (*Bipolarissorokiniana (Sacc.)* Shoemaker)

The resistance to infringement.

1 highly resistance(HR)

3 disease-resistant(R)

5 Zhong Kang (MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.7 Resistance to wood rot

The resistance of wheat plants to infection by the damping-off pathogen (*Rhizoctonia cerealis* Van der Hoeven)

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.8 Resistance to yellow dwarf disease

The resistance of barley plants to infection by the Barley yellow dwarf virus.

1 highly resistance(HR)

3 disease-resistant(R)

5 Zhong Kang (MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.9 Total dieback resistance

Wheat plants are susceptible to the whole-etch pathogen, Gaeumannomyces graminis (Sacc.) Arx. olivcr) the resistance to infringement.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.10 Aphid resistance

The resistance of wheat to damage caused by wheat aphids (wheat long tube aphid *Sitobion avenae (* Fabricius), wheat binary aphid *Schizaphis graminum (* Rondani), rice stem aphid *Rhopalosiphum padi (* Linnaeus), and wheat long tube aphid without web *Metopolophium dirhodum (* Walker))

1 highly resistance(HR)

3 resist(R)

5 Zhong Kang (MR)

7 feel(S)

9 High Sensitivity (HS)

8.11 Siphonate resistance

The resistance of wheat plants to damage caused by the sap-sucking insect Sitodiplosis *mosellana Genin* varies in intensity.

1 highly resistance(HR)

3 resist(R)

5 Zhong Kang (MR)

7 feel(S)

9 High Sensitivity (HS)

9 Other Characteristics and Features

9.1 Hybrid Wheat

The first generation produced through the cross between restorer lines and sterile lines, which exhibits heterosis.

1 Three-Line Hybrid Wheat

2 Two-line hybrid wheat

3 Chemical control of hybrid wheat

9.2 Non-aneuploid wheat

Wheat germplasm resources whose chromosome number deviates from its complete multiple of the base number—including individual plants or complete sets—contain, within their chromosome sets, one or more individual chromosomes or chromosome arms that exceed or fall short of the normal number.

l Primary aneuploidy

2 Secondary aneuploidy

9.3 Karyotype

Karyotype refers to the aggregate of chromosomal characteristics observed during metaphase of cell division; it is a formula representing the number, size, morphology, and structural features of the chromosomes.

9.4 Near-isogenic genetic lines

Near isogenic lines (NILs) refer to lines that, through a series of backcrossing processes, share an identical or closely related genetic background and differ only in specific chromosomal regions.

In breeding practice, a donor parent carrying a marker trait gene is crossed with a recurrent parent, followed by multiple backcrosses; in each generation, only individuals carrying the target gene are selected for backcrossing with the recurrent parent, thereby obtaining lines that share the same genetic background as the recurrent parent except for the target gene.

9.5 Recombinant Inbred Strain

Cross two varieties to produce the F1 generation; self-pollinate to obtain the F2 generation; then randomly select several hundred to several thousand individual plants from the F2 generation for self-pollination, planting only one seed per plant; continue this process through F6–F8 to generate several hundred recombinant inbred lines.

A population in which the offspring genome is relatively homozygous, obtained through multiple generations of selfing and single-seed transmission (Single Seed Descendant, abbreviated as SSD) from an F2 population, is referred to as an RIL (Recombinant Inbred Strain) population.

9.6 DHpopulation

By subjecting F1 plants to in vitro anther culture or inducing the production of haploid plants through specialized techniques, followed by chromosome doubling, a homozygous, permanent population is obtained; this is the doubled haploid population, commonly referred to as the DH population, which is frequently employed in QTL studies of wheat.

9.7 Fingerprinting and Molecular Markers

Molecular marker types and their characteristic parameters for wheat germplasm fingerprint profiles and important traits.

9.8 remarks

A detailed description of the specific descriptors or special codes for wheat germplasm.